

MAT 2001 - Statistics for Engineers.

Module 1: Introduction to Statistics. →

① Statistics are measurements, enumerations or estimates of natural phenomenon, usually systematically arranged, analysed & presented as to exhibit important inter-relationships among them."

A.M. Tuttle

② Statistics is the science of estimates & probabilities.

Booldington

③ Statistics is the branch of scientific method which deals with the data obtained by counting or measuring the properties of population of natural phenomena.

Kendall

④ Croxon & Cowden defines statistics as the science of collection, presentation, analysis & interpretation of numerical data.

Ex (1):

You want to start a new business & want to know the average household income per year in your primary business region.

Ex (2):

You are the part of anti-smoking campaign in your school. You are concerned about the general health of your fellow students. & want to know percentage of the students are regular smokers.

Ex 1

As a quality control expert, you want to know the proportion of good computer chips produced by manufacturing unit.

Ex 2

You are working as a consultant for a telephone company & want to find out the proportion of households in particular state having mobile phones.

Ex 3

University decides to increase the student fee from next academic year. The student union want to know what percentage of students support the fee hike.

Note that, each of the above examples, there is a collection of individuals or a groups of items under study. This is called population.

Population : A population is a collection of all distinct individuals or objects or items under study. The number of entities in a population, called the population size.

①. In Ex 1, Population is the collection of households in printing business, since we are interested in the average income of households.

②. In Ex 2, Population under study is the collection of enrolled students (in a particular semester) since you want to conduct your study on them.

③ In Ex ③, population is the collection of all computer chips produced by manufacturing unit. Then you want to know what percentage of this population is good.

④ In Ex ④, the population consists of all households in particular state. You want to determine the proportion of this population having mobile phones.

⑤ In Ex ⑤, population is the collection of all students enrolled in the current semester in university. (~~the~~ percent of total no. of students supporting tuition fee hike)

⇒ Variable → Once a population is fixed then we want to study a characteristic of individual (or objects or items) in the population. This individual characteristic is called a variable (value of which can vary from individual to individual, or from one object to another within the population).

The variable in

- ① Ex 1, is the income of households.
- ② Ex 2, is the smoking habit of student.
- ③ Ex 3, is the quality of computer chip.
- ④ Ex 4, is ownership of mob. phone(s) by household.
- ⑤ Ex 5, is the opinion of a student regarding fee hike.

Most of the variables can be classified (4) in two broad categories:

① Categorical (or qualitative) variable → It assigns a categorical (non-numerical) value to each individual or object.
Ex: gender, nationality, etc.

② Quantitative variable → It assigns a numerical value to each individual or object.
Ex: age, income, height, etc.

① In Ex ①, income of household is a ~~qualitative~~ quantitative variable.

② In Ex ②, smoking habit of student is a categorical variable (since a person is either a smoker or a non smoker).

③ In Ex ③, quality of chip is a categorical variable (since chip is ~~classified~~ as either good or bad).

④ In Ex ④, ownership of mob. phone is a ~~cat~~ categorical variable (since a household either owns or does not own).

⑤ In Ex ⑤, opinion of student is a categorical variable (since a student either favours or does not not favour).

Books: ① Probability & Statistics for engineers & scientists, R.E. Walpole, R.H. Myers, S.L. Myers & K. Ye, 9th Edition, Pearson Education, 2012

② Fundamental of Mathematical Statistics, S.C. Gupta, V.K. Kapoor, Sultan Chand & Sons.

⇒ Statistics → ① Introduction ② Data Science ③ Data Analysis ⑤

⇒ Frequency Distribution → ① Discrete Frequency distribution ② Continuous Frequency distribution

⇒ Measures of Central Tendency →

① Arithmetic mean or average.

② Median

③ Mode

④ Geometric mean

⑤ Harmonic mean.

①. Arithmetic mean or Average (M) → Arithmetic mean of a set of observations is their sum divided by the number of observations, e.g. the A.M. $\bar{x}$ of n observations $x_1, x_2, \dots, x_n$ is given by

$$\bar{x} = \frac{1}{n} (x_1 + x_2 + \dots + x_n) = \frac{1}{n} \sum_{i=1}^n x_i$$

In case of frequency distribution $x_i / f_i, i=1, 2, \dots, n$, where f_i is the frequency of the variable x_i

$$\bar{x} = \frac{f_1 x_1 + f_2 x_2 + \dots + f_n x_n}{f_1 + f_2 + \dots + f_n} = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i}$$

$$\bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i \quad \left[\sum_{i=1}^n f_i = N \right]$$

In case of grouped or continuous frequency distribution, x is taken as the mid-value of corresponding class.

Arithmetic mean or average →

$x_i: x_1 \quad x_2 \quad \dots \quad x_n$
 $f_i: f_1 \quad f_2 \quad \dots \quad f_n$

$$\text{Mean} = M = \frac{\sum_{i=1}^n f_i x_i}{\sum_{i=1}^n f_i} = \frac{1}{N} \sum_{i=1}^n f_i x_i, \text{ where } N = \sum_{i=1}^n f_i$$

Ex: Find the arithmetic mean of the following frequency distribution: ⑥

x :	1	2	3	4	5	6	7
f :	5	9	12	17	14	10	6

Sol:

x	f	fx
1	5	5
2	9	18
3	12	36
4	17	68
5	14	70
6	10	60
7	6	42
	73	299

$$\bar{x} = \frac{1}{N} \sum fx = \frac{1}{73} \cdot 299 = 4.09.$$

Ex: Calculate the A.M. of the marks from the following ~~table~~ table:

Marks:	0-10	10-20	20-30	30-40	40-50	50-60
No. of students:	12	18	27	20	17	6

Sol:

Marks	No. of Students (f)	Mid-point (x)	fx
0-10	12	5	60
10-20	18	15	270
20-30	27	25	675
30-40	20	35	700
40-50	17	45	765
50-60	6	55	330
Total	100		2800

$$A.M. \text{ or } \bar{x} = \frac{1}{N} \sum f_i x_i = \frac{1}{100} 2800 = 28. \quad (7)$$

Arithmetic mean or average → Note that if the values of x or (and) f are large, the calculation of mean by formula is quite time-consuming & tedious. The arithmetic is reduced to a great extent by taking the deviations of the given values from any arbitrary point 'A' as explained below.

$$\text{Let } d_i = x_i - A \text{ then } f_i d_i = f_i (x_i - A) \\ f_i d_i = f_i x_i - A f_i$$

$$\text{Summing both sides over } i \left. \begin{array}{l} \text{from 1 to } n, \text{ we get} \end{array} \right\} \sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - A \sum_{i=1}^n f_i$$

$$\sum_{i=1}^n f_i d_i = \sum_{i=1}^n f_i x_i - A \cdot N$$

$$\Rightarrow \frac{1}{N} \sum_{i=1}^n f_i d_i = \frac{1}{N} \sum_{i=1}^n f_i x_i - A$$

$$\frac{1}{N} \sum_{i=1}^n f_i d_i = \bar{x} - A$$

where $\bar{x}$ is the A.M. of the distribution

$$\therefore \boxed{\bar{x} = A + \frac{1}{N} \sum_{i=1}^n f_i d_i}$$

Notes In case of grouped or continuous frequency distribution, the arithmetic is reduced to a still greater extent by taking

$d_i = \frac{x_i - A}{h}$, where A is an arbitrary point & h is the common magnitude of class interval.

~~The~~ In this case, we have

$$h d_i = x_i - A$$

& proceeding exactly similarly as above, we get

$$\boxed{\bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i}$$

Ex) Calculate the mean for the following frequency distribution: ⑧.

Class interval: 0-8 8-16 16-24 24-32 32-40 40-48
 Frequency : 8 7 16 24 15 7

Sol:

Class Interval	mid-value (x)	Frequency (f)	$d = \frac{x-A}{h}$	fd
0-8	4	8	-3	-24
8-16	12	7	-2	-14
16-24	20	16	-1	-16
24-32	28	24	0	0
32-40	36	15	1	15
40-48	44	7	2	14
		77		-25

Here we take $A=28$ & $h=8$. (common magnitude of class interval)

$$\therefore \bar{x} = A + \frac{\sum fd}{N} = 28 + \frac{8 \times (-25)}{77}$$

$$= 28 - \frac{200}{77} = 25.404$$

2. Median (M_d) $\rightarrow$ Median of a distribution is the value of the variable which divides it into two equal parts. It is the value which exceeds & is exceeded by the same no. of observations, i.e. it is the value s.t. the no. of observations above it is equal to the no. of observations below it. The median is thus a positional average.

In case of ungrouped data, if the no. of observations is odd then median is the middle value after the values have been arranged in ascending or descending order of magnitude.

In case of even no. of observations, there are two middle terms & median is obtained by taking the arithmetic mean of the middle terms.

Ex $\rightarrow$ The median of the values 25, 20, 15, 35, 18 i.e. 15, 18, 20, 25, 35 is 20 & the median of 8, 20, 50, 25, 15, 30 i.e. of 8, 15, 20, 25, 30, 50 is $\frac{1}{2}(20+25) = 22.5$.

$\Rightarrow$ Median for discrete frequency distribution $\rightarrow$ In case of discrete frequency distribution median is obtained by considering the cumulative frequencies. The steps for calculating median are given below:

- ① Find $N/2$, where $N = \sum f_i$
- ② See the (less than) cumulative frequency (C.f.) just greater than $N/2$.
- ③ The corresponding value of x is median.

$\Rightarrow$ Ex: Obtain the median for the following frequency distribution:

x :	1	2	3	4	5	6	7	8	9
f :	8	10	11	16	20	25	15	9	6

Soln

x	f	c.f.
1	8	8
2	10	18
3	11	29
4	16	45
5	20	65
6	25	90
7	15	105
8	9	114
9	6	120
	<u>120</u>	

c.f. just greater than $\frac{N}{2}$

Hence $N=120 \Rightarrow \frac{N}{2} = 60$.

C.F = 65

Cumulative frequency (C.f.) just greater than $N/2$, is 65 & the value of x corresponding to 65, Therefore median is 5.

$\Rightarrow$ Median for continuous frequency distribution $\rightarrow$ In the case of continuous frequency distribution, the class corresponding to the c.f. just greater than $N/2$ is called the median class & the value median is obtained by the following formula:

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right)$$

where

l is the lower limit of the median class,

f is the frequency of median class.

h is the magnitude of median class,

c is the c.f. of the class preceding the median class & $N = \sum f$.

Ex Find the median wage of the following distribution

Wage (in Rs)	20-30	30-40	40-50	50-60	60-70
Nb. of labourers	3	5	20	10	5

<u>Solⁿ</u>	Wages (in Rs.)	No. of labourers	C.f.
	20-30	3	3
	30-40	5	8
	40-50	20	<u>28</u>
	50-60	10	38
	60-70	5	43

Here $\frac{N}{2} = \frac{43}{2} = 21.5$

$(CF > \frac{N}{2})$ just greater than

Cumulative frequency just greater than 21.5 is 28 & the corresponding class is 40-50. The median class is 40-50.

$$\text{Median} = 40 + \frac{10}{20} (21.5 - 8) = 40 + 6.75 = 46.75$$

Thus median wage is Rs 46.75.

③ Mode (Mo) → Mode is the value which occurs most frequently in a set of observations & around which the other items of the set cluster densely.
In other words, mode is the value of the variable which is predominant in the series.

Mode for discrete frequency distribution → In case of discrete frequency distribution, mode is the value of x corresponding to maximum frequency. (42)

Ex: For the following discrete frequency distribution,

x	1	2	3	4	5	6	7	8
f	4	9	16	25	22	15	7	3

the value of x corresponding to the maximum frequency, viz 25 is 4. Hence mode is 4.

⇒ Mode for discrete frequency distribution

But in any one (or more) of the following cases:

- ① if the max. frequency is repeated.
- ② if the max. frequency occurs in the very beginning or at the end of the distribution, &
- ③ if there are irregularities in the distribution, the value of mode is determined by the method of grouping which is illustrated below by an example.

⇒ Method for discrete frequency distribution

Method of Grouping →

Ex: Find the mode of the following frequency distribution:

Size (x)	1	2	3	4	5	6	7	8	9	10	11	12
Frequency (f)	3	8	15	23	35	40	32	28	20	45	14	6

Solⁿ (i) Combine frequencies 2 by 2, Leave 1st frequency & combine frequencies 2 by 2 we get (iii) (13)
 leave 1st frequency & combine frequencies 3 by 3 we get (iv). Leave 1st frequency &
 combine 3-3 we get (v). Leave 1st frequency & combine 3-3 we get (vi).

Size (n)	(i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11	23	26		
2	8					
3	15	38	58		46	
4	23			98		73
5	35	75				
6	40		72		107	
7	32	60		80		100
8	28		48			
9	20	65			93	
10	45		59	65		79
11	14	20				
12	6					

Analysis Table

Column no. (1)	Maximum Frequency (2)	Value of combination of values of x giving max frequency in (2), (3)
(i)	45	10
(ii)	75	5, 6
(iii)	72	6, 7
(iv)	98	4, 5, 6
(v)	107	5, 6, 7
(vi)	100	6, 7, 8

On examining the values in column (3) above, we find that the value 6 is repeated the max. no. of times & hence the value of mode is 6 & not 10 which is an irregularity.

$$\text{Mode} = 6$$

⇒ Mode for continuous frequency distribution →

$$\text{Mode} = l + \frac{h(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)} = l + \frac{h(f_1 - f_0)}{2f_1 - f_0 - f_2}$$

where l is lower limit of ~~modal class~~ modal class.

h is magnitude of modal class.

f_1 is frequency of modal class.

f_0 & f_2 are frequencies of the classes preceding & succeeding the modal class respectively.

Q Find the mode for the following distribution:

Class interval :	0-10	10-20	20-30	30-40	40-50	50-60
Frequency :	5	8	7	12	28	20

Solⁿ Max. frequency is 28. Thus the class 40-50 is modal class.

$$\begin{aligned} \text{Mode} &= 40 + \frac{10(28 - 12)}{2 \times 28 - 12 - 20} = 40 + 6.666 \\ &= 46.67 \text{ (approx.)} \end{aligned}$$

Geometric Mean (G) is G.M. of a set of n observations is the n th root of their product. Thus G.M. G of n observations $x_i; i=1, 2, \dots, n$ is

$$G = (x_1 \cdot \dots \cdot x_n)^{1/n}$$

Taking log on both sides, we get

$$\log G = \frac{1}{n} [\log x_1 + \dots + \log x_n]$$

$$\log G = \frac{1}{n} \sum_{i=1}^n \log x_i$$

$$\therefore G = \text{Antilog} \left[\frac{1}{n} \sum_{i=1}^n \log x_i \right]$$

In case of frequency distribution $x_i / f_i, (i=1, 2, \dots, n)$.

G.M. G is given by:

$$G = \left[x_1^{f_1} \cdot x_2^{f_2} \cdot \dots \cdot x_n^{f_n} \right]^{1/N}, \text{ where } N = \sum_{i=1}^n f_i$$

Taking log on both sides, we get

$$\log G = \frac{1}{N} [f_1 \log x_1 + f_2 \log x_2 + \dots + f_n \log x_n]$$

$$\log G = \frac{1}{N} \sum_{i=1}^n f_i \log x_i$$

$$G = \text{Antilog} \left[\frac{1}{N} \sum_{i=1}^n f_i \log x_i \right]$$

In case of grouped or continuous frequency distribution, x is taken to be the value corresponding to the midpoint of the class intervals.

Q. The geometric mean of 10 observations on a certain variable was calculated ~~as~~ as 16.2.

It was later discovered that one of the observations was wrongly recorded as 12.9 in fact it was 21.9. Apply appropriate correction & calculate the correct G.M.

Solⁿ G.M. of n observations $x_1, x_2, \dots, x_n$

is given by

$$G = (x_1 \cdot \dots \cdot x_n)^{1/n}$$

If x_1 is the observation copied wrongly instead of correct value x_1' , then the corrected G.M. G' is given by:

$$G' = (x_1' x_2 \dots x_n)^{1/n} = \left(\frac{x_1'}{x_1} \cdot x_1 x_2 \dots x_n \right)^{1/n}$$

$$G' = (x_1 x_2 \dots x_n)^{1/n} \left(\frac{x_1'}{x_1} \right)^{1/n} = G \left(\frac{x_1'}{x_1} \right)^{1/n}$$

$$\text{Given } G = 16.2, n = 10, x_1 = 12.9, x_1' = 21.9$$

$$\text{Corrected G.M. } G' = 16.2 \left(\frac{21.9}{12.9} \right)^{1/10}$$

$$\log_{10} G' = \log_{10} 16.2 + \frac{1}{10} (\log_{10} 21.9 - \log_{10} 12.9)$$

$$= 1.2095 + \frac{1}{10} (1.3404 - 1.1106)$$

$$= 1.23248$$

$$G' = \text{Antilog}(1.2325) = 17.08$$

Harmonic Mean $\rightarrow$ H.M. of a no. of observations (16)

is the reciprocal of the arithmetic mean of the reciprocals of the given values. Thus, H.M. H of n observations $x_i, i=1, 2, \dots, n$ is

$$H = \frac{1}{\frac{1}{n} \sum_{i=1}^n \left(\frac{1}{x_i}\right)}$$

In case of frequency distribution $x_i / f_i; (i=1, 2, \dots, n)$

$$H = \frac{1}{\frac{1}{N} \sum_{i=1}^n (f_i / x_i)}, \text{ where } N = \sum_{i=1}^n f_i$$

$\Rightarrow$ Measures of Variability or Measures of Dispersion $\rightarrow$

Dispersion - Variations or Scatteredness.

List of Measures of Dispersion:

① Range $\rightarrow$ Range = Max. value - Min. value.

The range is the difference b/w two extreme observations of the distribution. If A & B are the greatest & smallest observations respectively in a distribution, then its range is $A - B$. It is not at all a reliable measure of dispersion (in skewed & bimodal only).

② Quantile Deviation or Semi-interquartile Range $\rightarrow$
Partition Values $\rightarrow$ Those are the values which divide the series into a number of equal parts.

Quantiles $\rightarrow$ The three points which divide the series into four equal parts are called quantiles. The I^{st} , II^{nd} & III^{rd} points are known as the I^{st} , II^{nd} & III^{rd} quantiles respectively.

The 1st quantile Q_1 is the value which exceed 25% of the observations & is exceeded by 75% of the observations. The 2nd quantile, Q_2 , coincides with median. The 3rd quantile Q_3 is the point which has 75% observations before it & 25% observation after it. (17)

Deciles & Percentiles) The nine points which divide the series into ten equal parts are called deciles whereas percentiles are the ninety-nine points which divide the series into hundred equal parts.

Quantile Deviation or Semi-interquartile Range (Q)

$$Q = \frac{1}{2} (Q_3 - Q_1)$$

where Q_1 & Q_3 are the 1st & 3rd quantiles of the distribution respectively.

Quantile deviation is definitely a better measure than the range as it makes use of 50% of the data. But since it ignores the other 50% of the data, it can not be regarded as a reliable measure.

Q1 Eight coins were tossed together & no. of heads resulting were noted. The operation was repeated 256 times & the frequencies (f) that were obtained for different values of x , the no. of heads, are shown in table. Calculate median, quantiles.

x :	0	1	2	3	4	5	6	7	8
f :	1	9	26	59	72	52	29	7	1

∴ Mean deviation is based on all the observations. It is better ~~than~~ measure of dispersion than range or quartile deviation. But the steps of ignoring the sign of deviation $(x_i - A)$ creates artificiality & renders it useless for mathematical treatment. Note that mean deviation is least when taken from median.

④ Standard deviation (σ) → S.D. is the +ve square root of the arithmetic mean of the squares of the deviations of the given values from their A.M. For the arithmetic distribution x_i/f_i , $i=1, 2, \dots, n$.

$$\sigma = \sqrt{\frac{1}{N} \sum f_i (x_i - \bar{x})^2}$$

where $\bar{x}$ is A.M. of distribution & $N = \sum f_i$.

Variance → The square of S.D. is called the variance & is given by:

$$\sigma^2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2$$

Root mean square deviation (δ) →

$$\delta = \sqrt{\frac{1}{N} \sum_i f_i (x_i - A)^2}$$

where A is any arbitrary number

→ any number.

δ^2 is called mean square deviation.

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 ⇒ Difference formulae for calculating variance ⇒

(20)

$$\sigma^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$

More precisely we write it as σ_x^2 ,

$$\sigma_x^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$

$$= \frac{1}{N} \sum f_i (x_i^2 + \bar{x}^2 - 2x_i \bar{x})$$

$$= \frac{1}{N} \sum f_i x_i^2 + \frac{\bar{x}^2}{N} \sum f_i - 2\bar{x} \cdot \frac{1}{N} \sum f_i x_i$$

$$= \frac{1}{N} \sum f_i x_i^2 + \bar{x}^2 - 2\bar{x}^2 \quad \left\{ \begin{array}{l} \text{for frequency} \\ \text{dis'n} \\ \bar{x} = \frac{1}{N} \sum f_i x_i \end{array} \right.$$

$$\sigma_x^2 = \frac{1}{N} \sum f_i x_i^2 - \left(\frac{1}{N} \sum f_i x_i \right)^2 \quad \left\{ \begin{array}{l} \text{For loose value} \\ \text{of } x \end{array} \right.$$

We know that if $d_i = x_i - A$, then

$$\bar{x} = A + \frac{1}{N} \sum f_i d_i$$

A is arbitrary pt.

∴ $A - \bar{x} = -\frac{1}{N} \sum f_i d_i$

Hence

$$\sigma_d^2 = \frac{1}{N} \sum f_i d_i^2 + \left(-\frac{1}{N} \sum f_i d_i \right)^2 + 2 \left(-\frac{1}{N} \sum f_i d_i \right) \left(\frac{1}{N} \sum f_i d_i \right)$$

$$= \frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2$$

$$\sigma_x^2 = \sigma_d^2$$

Let $d_i = x_i - A$ — (1)

Multiply by f_i , summing over i from 1 to n & dividing by N , we get

$$\frac{1}{N} \sum f_i d_i = \frac{1}{N} \sum f_i x_i - A \cdot \frac{1}{N} \sum f_i$$

$$\bar{d} = \bar{x} - A \quad \text{--- (2)}$$

① - ②, we get

$$\therefore \sigma_x^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \frac{1}{N} \sum f_i (d_i - \bar{d})^2 = \sigma_d^2$$

Hence variance & consequently S.D. is independent of change of origin - for continuous frequency distribution

Now If $d_i = \frac{x_i - A}{h}$, then $x_i = A + h d_i$ (1)

$$\therefore \bar{x} = A + h \cdot \frac{1}{N} \sum f_i d_i = A + h \bar{d} \quad (2)$$

$$(1) - (2), \text{ we get } x_i - \bar{x} = h(d_i - \bar{d})$$

$$\therefore \sigma_x^2 = \frac{1}{N} \sum f_i (x_i - \bar{x})^2$$

$$= h^2 \cdot \frac{1}{N} \sum f_i (d_i - \bar{d})^2$$

$$\sigma_x^2 = h^2 \sigma_d^2 \quad \sigma_d^2 = \left[\frac{1}{N} \sum f_i d_i^2 - \left(\frac{1}{N} \sum f_i d_i \right)^2 \right]$$

Which shows that variance is independent of change of origin but not of scale.

Coefficient of dispersion $\rightarrow$ Whenever we want to compare the variability of two series which differ widely in their averages or which are measured in different units, we do not merely calculate the measures of dispersion but we calculate the coefficient of dispersion which are pure numbers independent of units of measurement. The coefficient of dispersion (C.D.) based on different measures of dispersion are as follows:

① C.D. based upon range = $\frac{A - B}{A + B}$, where A & B are the greatest & smallest items in series.

② Based quartile deviation:

$$CD = \frac{(Q_3 - Q_1)/2}{(Q_3 + Q_1)/2} = \frac{Q_3 - Q_1}{Q_3 + Q_1}$$

(3) Based upon mean deviation (28)

$$C.D. = \frac{\text{Mean deviation}}{\text{Average from which it is calculated}}$$

(4) Based upon standard deviation:

$$C.D. = \frac{S.D.}{\text{Mean}} = \frac{\sigma}{\bar{x}}$$

Coefficient of variation → 100 times the coefficient of dispersion based upon S.D. is called coeff. of variation (C.V.).

$$C.V. = 100 \times \frac{\sigma}{\bar{x}}$$

Scanned by 2.38

Q. Calculate the mean & S.D. for the following table giving the age distribution of 542 members.

Age in years:	20-30	30-40	40-50	50-60	60-70	70-80
No. of members:	3	61	132	153	140	51

Also find C.D. & C.V. 80-90
2

Solⁿ ∴ For the large values of frequency distribution.

we take $d_i = \frac{x - A}{h}$, where A is arbitrary pt. & h is common magnitude of class interval.

$$\bar{x} = A + \frac{h}{N} \sum_{i=1}^n f_i d_i$$

Here

Age Group	Mid value (x)	Frequency (f)	$d = \frac{x - 55}{10}$	fd	fd^2 (23)
20-30	25	3	-3	-9	27
30-40	35	61	-2	-122	244
40-50	45	132	-1	-132	132
50-60	55	153	0	0	0
60-70	65	140	1	140	140
70-80	75	51	2	102	204
80-90	85	2	3	6	18
		$N = \sum f = 542$			
			$\sum fd = -15$		$\sum fd^2 = 765$

$$\bar{x} = A + h \frac{\sum fd}{N} = 55 + \frac{10 \times (-15)}{542} = 55 - 0.28$$

$$\bar{x} = 54.72 \text{ years}$$

$$\sigma^2 = h^2 \left[\frac{1}{N} \sum fd^2 - \left(\frac{1}{N} \sum fd \right)^2 \right]$$

$$= 100 \left[\frac{765}{542} - (0.028)^2 \right]$$

$$= 100 \times 1.4107$$

$$= 141.07$$

$$\therefore \sigma (\text{standard deviation}) = \sqrt{141.07}$$

$$= 11.88 \text{ years}$$

$$CD = \frac{\sigma}{\bar{x}}$$

$$CV = 100 \times \frac{\sigma}{\bar{x}} =$$

2.9.9 SCG
Moments The r^{th} moment of variable x about any point $x=A$, usually denoted by μ_r' is given by

$$\mu_r' = \frac{1}{N} \sum_i f_i (x_i - A)^r, \quad \sum_i f_i = N$$

$$= \frac{1}{N} \sum_i f_i d_i^r, \quad \text{where } d_i = x_i - A.$$

The r^{th} moment of variable x about the mean $\bar{x}$, usually denoted by μ_r is given by:

$$\mu_r = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^r = \frac{1}{N} \sum_i f_i z_i^r$$

where $z_i = x_i - \bar{x}$.

Particular Cases → In particular

$$\mu_0 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^0 = \frac{1}{N} \sum_i f_i = 1$$

& $\mu_1 = \frac{1}{N} \sum_i f_i (x_i - \bar{x}) = 0$, being the algebraic sum of deviations from the mean.

Also $\mu_2 = \frac{1}{N} \sum_i f_i (x_i - \bar{x})^2 = \sigma^2$.

These results, viz, $\mu_0=1$, $\mu_1=0$ & $\mu_2=\sigma^2$ are of fundamental importance & should be committed to memory.

Pearson's Coefficients → Karl Pearson defined the following four coefficients, based upon the 1st four moments about mean:

$$\beta_1 = \frac{\mu_3^2}{\mu_2^3}, \quad \gamma_1 = +\sqrt{\beta_1} \quad \& \quad \beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3$$

Skewness $\rightarrow$ Literally, skewness means 'lack of symmetry'. We study skewness to have an idea about the shape of the curve which can draw with the help of given data. A distribution is said to be skewed if

- (i) Mean, median & mode fall at different points, i.e. $\text{mean} \neq \text{median} \neq \text{mode}$.
- (ii) Quartiles are not equidistant from median, &
- (iii) The curve drawn with the help of given data is not symmetrical but stretched more to one side than to the other.

Measures of skewness $\rightarrow$ Various measures of skewness are:

① $S_k = M - M_d$ ② $S_k = M - M_0$,

where M is the mean,
 M_d is the median
& M_0 is the mode of distribution.

③ $S_k = (Q_3 - M_d) - (M_d - Q_1)$.

These are absolute measures of skewness.

As in dispersion, for comparing two series we don't calculate these absolute measures but we calculate the relative measures called the coefficients of skewness, which are pure numbers independent of units of measurements.

$\rightarrow$ Coefficients of Skewness $\rightarrow$

① Prof. Karl Pearson's Coefficient of Skewness:

$$S_k = \frac{(M - M_0)}{\sigma}$$

② Prof. Bowley's Coefficient of Skewness. Based on quartiles: (26)

$$S_k = \frac{(Q_3 - M_d) - (M_d - Q_1)}{(Q_3 - M_d) + (M_d - Q_1)} = \frac{Q_3 + Q_1 - 2M_d}{Q_3 - Q_1}$$

③ Based upon moments, coefficient of skewness is

$$S_k = \frac{\sqrt{\beta_1} (\beta_2 + 3)}{2(5\beta_2 - 6\beta_1 + 9)} \quad \text{--- Vesign}$$

where symbols have their usual meaning.

Thus $S_k = 0$ if either $\beta_1 = 0$ or $\beta_2 = -3$.

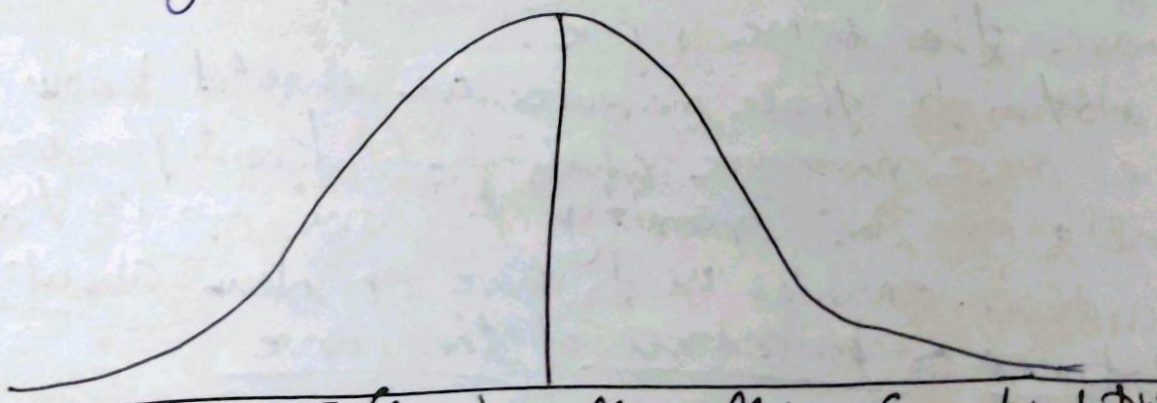
But since $\beta_2 = \frac{\mu_4}{\mu_2^2}$ can not be negative,

$$S_k = 0 \text{ iff } \beta_1 = 0.$$

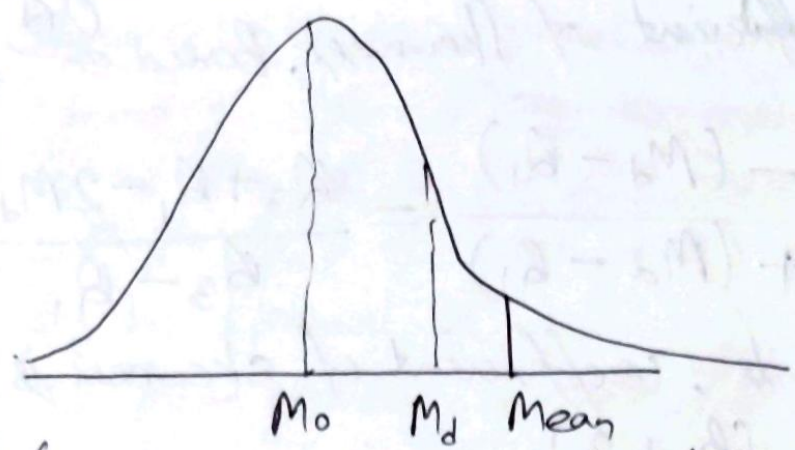
Thus for a symmetrical distribution $\beta_1 = 0$.

In this respect β_1 is taken to be a measure of skewness.

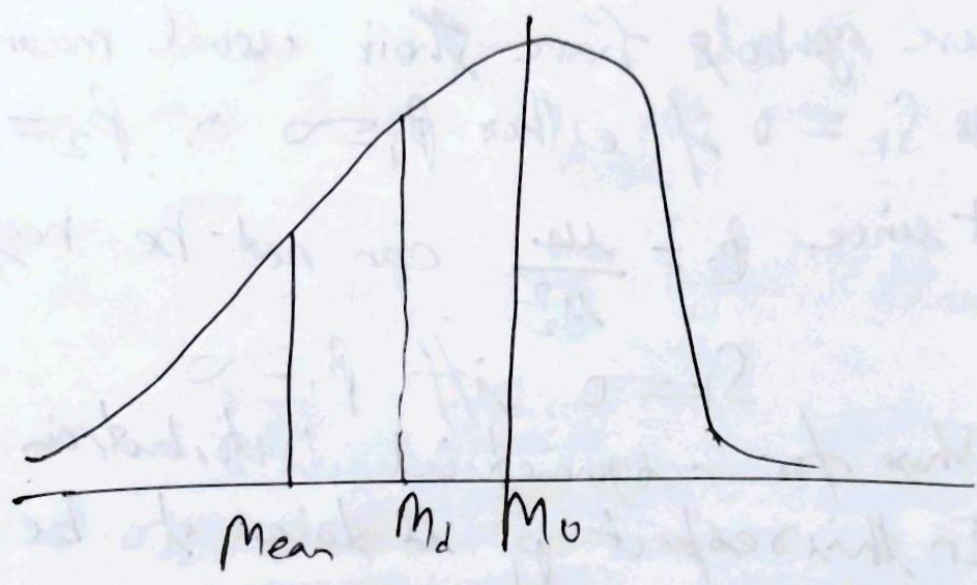
The skewness is +ve if the longer tail of the distribution lies towards the higher values of the variates (the right) i.e. if the curve drawn with the help of given data is stretched more to the right than to the left & is negative in the contrary case.



$\bar{x} (\text{Mean}) = M_0 = M_d$. Symmetrical Distribution



(+Vely Skewed Distribution)



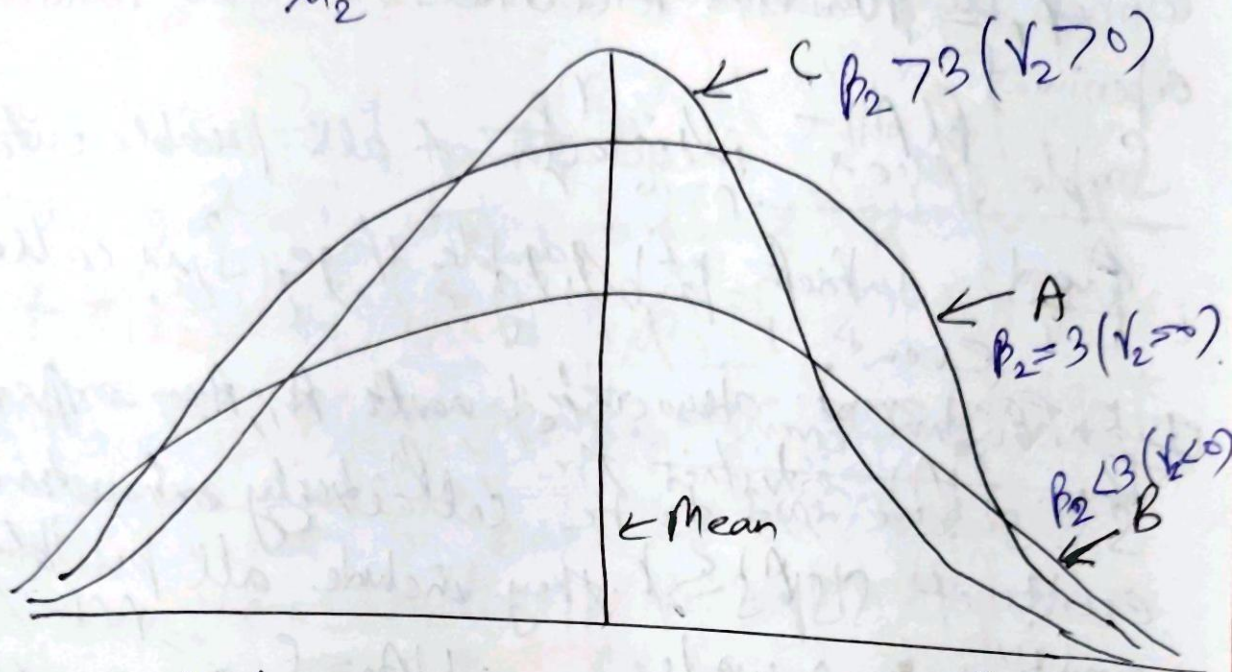
(-Vely Skewed Distribution)

Kurtosis If we know the measures of central tendency Dispersion & Skewness, we still cannot form a complete idea about the distribution as will be clear from the following figure in which all these curves A, B & C are symmetrical about the mean mean 'm' & have the same range.

In addition to these measures, we should know one more measure which Prof Karl Pearson calls as the 'Convexity of Curve' or Kurtosis. Kurtosis enables us to have an idea about the flatness or peakedness of the curve.

It is measured by the coefficient β_2 or (28)
its deviation γ_2 given by

$$\beta_2 = \frac{\mu_4}{\mu_2^2}, \quad \gamma_2 = \beta_2 - 3.$$



Curve of type 'A' which is neither flat nor peaked is called the normal curve or mesokurtic curve & for such a curve $\beta_2 = 3$ i.e. $\gamma_2 = 0$. Curve of type 'B' which is flatter than the normal curve is known as platykurtic curve & for such a curve $\beta_2 < 3$ i.e. $\gamma_2 < 0$. Curve of type 'C' which is more peaked than the normal curve is called leptokurtic curve & for such a curve $\beta_2 > 3$ i.e. $\gamma_2 > 0$.